ESE 4011 Question Map

Every question from the 2022 term papers, grouped by topic and ranked by how often it comes back and how many marks it carries. Answers are in the ESE 4011 Study Pack (HTML and PDF) next to this file.

24-hour list | Pass plan | Frequency grid | Section A | Section B

24-hour priority list

You answer **3 of 4 questions in each part**, so each part has its own list. Study down both lists until the green **pass line**: about 4 h for Part A and 4 h for Part B, which leaves time for sleep and a last read of the one-shot notes. Alternate between parts (A block 1, B block 1, A block 2 …) so neither is left short. **Chance**: how often the topic came up, newer papers weigh more (2022 ×1). **Covered**: share of that part's past-paper marks you have studied once you finish the row.

Only one paper (2022) exists, so every topic has the same chance; the plan therefore puts easy topics first. The fuel cell block (A7–A12) is one big family: learn the comparison table in the A10 memory box and you can answer most of Section A Q3 and Q4. The two numericals (B2 capacitors, B9 XRD) are quick sure marks with the calculators.

Section A: Solar cell materials and fuel cells

Block 1: very likely and easy · 3.8 h

Topic	Chance	Level	Covered
A8 Fuel cell stack (bipolar plates, series connection); DMFC layer structure	Very likely 100%	Easy	9%
A11 Phosphoric acid fuel cell (PAFC): electrolyte properties and functions, electrodes and catalyst	Very likely 100%	Easy	19%
A2 Crystalline silicon structures (mono-, multi-, amorphous); diffusion and drift device concepts	Very likely 100%	Easy	27%

A5 Chalcopyrite materials (CIS, CIGS) and the CdTe cell: layer structure and working principle	Very likely 100%	Easy	36%
A9 Role of PTFE in alkaline fuel cells; functions of bipolar plates in PEMFCs	Very likely 100%	Easy	44%
A12 Solid oxide fuel cell (SOFC): favourable characteristics, electrolyte types	Very likely 100%	Easy	53%
A4 A third-generation thin-film cell: dye-sensitised solar cell (working principle)	Very likely 100%	Easy	60%
A7 Fuel cell: definition, difference from a battery, working principle	Very likely 100%	Easy	67%
A10 Molten carbonate fuel cell (MCFC): construction, principle, features	Very likely 100%	Easy	74%

Block 2: very likely, medium effort · 2.2 h

Topic	Chance	Level	Covered
A3 High-efficiency concepts of c-Si cells; the PERL cell	Very likely 100%	Medium	84%
✔ Pass line: 84% of Section A marks covered, enough to choose 3 of the 4 questions (about 4.5 h)			
A6 Multijunction cells: challenges; spectral conversion	Very likely 100%	Medium	93%
A1 Losses in solar cells; optical losses	Very likely 100%	Medium	100%

Section B: Supercapacitors, hydrogen storage, thin films, XRD and electron microscopy

Block 1: very likely and easy · 2.9 h

Topic	Chance	Level	Covered
B10 SEM: backscattered vs secondary electrons; how SEM gives high resolution and depth of field	Very likely 100%	Easy	13%
B6 Chemical vapour deposition (CVD): process and role in thin films, powders, single crystals; PVD vs CVD	Very likely 100%	Easy	24%
B5 Sputtering: definition and process	Very likely 100%	Easy	31%
B8 Geometry and components of an X-ray diffractometer	Very likely 100%	Easy	37%
B7 Obtaining monochromatic Kα radiation from the X-ray tube spectrum	Very likely 100%	Easy	43%
B3 Pseudocapacitor vs double-layer capacitor	Very likely 100%	Easy	48%
B2 Numerical: capacitors 15 F and 8 F in series and in parallel across 50 V	Very likely 100%	Easy	52%

Block 2: very likely, medium effort · 3.0 h

Topic	Chance	Level	Covered
B4 Hydrogen storage in carbonaceous materials and CNTs: adsorption mechanism, key parameters, applications	Very likely 100%	Medium	66%
B9 XRD numericals: 2θ of Co (111) with Cu Kα; structure factor of an FCC cell	Very likely 100%	Medium	79%
✔ Pass line: 79% of Section B marks covered, enough to choose 3 of the 4 questions (about 4.4 h)			
B11 TEM: diffraction pattern and image together; preparing a 0.1 mm ceramic sample	Very likely 100%	Medium	91%
B1 Electric double-layer capacitor (EDLC): why high capacitance, fundamental principle	Very likely 100%	Medium	100%

Pass plan

Only the 2022 term final paper is available in the question bank (4 questions per section; each question has 3-4 parts). Section A = solar cell materials (losses, c-Si structures, PERL, thin films, DSSC, CdTe/CIGS, multijunction, spectral conversion) and fuel cells (basics, stack, DMFC, AFC/PEMFC parts, MCFC, PAFC, SOFC). Section B = supercapacitors (EDLC, pseudocapacitor, numerical), hydrogen storage in carbon/CNT, thin-film deposition (sputtering, CVD, PVD vs CVD), X-ray diffraction (Kα, diffractometer, Bragg and structure factor numericals) and electron microscopy (SEM, TEM). There is no teacher material folder for this course, so the answers follow the standard textbooks (Smets et al., "Solar Energy"; Larminie and Dicks, "Fuel Cell Systems Explained"; Conway, "Electrochemical Supercapacitors"; Cullity, "Elements of X-Ray Diffraction"; Goldstein/Williams and Carter for SEM/TEM). All diagrams were drawn for this pack. With one paper only, every question of 2022 is covered; expect the same topics to return with changed wording.

SECTION A: SOLAR CELL MATERIALS AND FUEL CELLS

12 topics

cover 100% of all Section A marks asked

SECTION B: SUPERCAPACITORS, HYDROGEN STORAGE, THIN FILMS, XRD AND ELECTRON MICROSCOPY

10 topics

cover 96% of all Section B marks asked

- Solar cell losses, c-Si structures, diffusion and drift devices, PERL
- Thin films: DSSC, chalcopyrite and CdTe; multijunction and spectral conversion
- Fuel cell basics vs battery, stacks, DMFC, PTFE in AFC, bipolar plates
- High-temperature and acid cells: MCFC, PAFC, SOFC

- EDLC, capacitor numerical, pseudocapacitor
- Hydrogen storage in carbon and CNTs; sputtering, CVD, PVD vs CVD
- Kα radiation, diffractometer, Bragg and structure factor numericals
- SEM (SE vs BSE), TEM modes and ceramic sample preparation

- **Section A is half solar, half fuel cells:** Q1-Q2 solar cells (losses A1, c-Si and diffusion/drift A2, PERL A3, DSSC A4, CdTe/CIGS A5, multijunction and spectral conversion A6); Q3-Q4 fuel cells (basics A7, stack/DMFC A8, PTFE and bipolar plates A9, MCFC A10, PAFC A11, SOFC A12).
- **Fuel cells are the safest marks:** learn the one table "type - electrolyte - ion - temperature - catalyst" (in the notes) and the anode/cathode reactions; it answers A7-A12.
- **Section B has the only numericals:** capacitors in series/parallel (B2, 6 marks) and XRD Bragg angle + FCC structure factor (B9, 18 marks). Each is a short step table: practise once.
- **Always draw:** optical losses, PERL cross-section, DSSC, CdTe stack, multijunction, fuel cell principle, stack, DMFC, MCFC, SOFC, EDLC, sputtering, CVD, Kα filter spectrum, diffractometer, SEM column, TEM ray diagrams.
- **Comparison tables score well:** fuel cell vs battery, EDLC vs pseudocapacitor, PVD vs CVD, SE vs BSE, imaging vs diffraction mode.

Frequency grid

Marks asked from each topic in each paper. Darker means more marks. Click a topic to jump to its questions.

Marks per paper: 1–8 9–16 17–24 25+

Sort Section A: Best order (likely + easy) | Most likely first | Easiest first | Paper order

SECTION A: SOLAR CELL MATERIALS AND FUEL CELLS: MARKS ASKED PER PAPER

Topic	2022	Years	Marks	Priority

A8 Fuel cell stack (bipolar plates, series connection); DMFC layer structure	13	1/1	13	Must study	Very likely · 100% Easy
A11 Phosphoric acid fuel cell (PAFC): electrolyte properties and functions, electrodes and catalyst	13	1/1	13	Must study	Very likely · 100% Easy
A2 Crystalline silicon structures (mono-, multi-, amorphous); diffusion and drift device concepts	12	1/1	12	Must study	Very likely · 100% Easy
A5 Chalcopyrite materials (CIS, CIGS) and the CdTe cell: layer structure and working principle	12	1/1	12	Must study	Very likely · 100% Easy
A9 Role of PTFE in alkaline fuel cells; functions of bipolar plates in PEMFCs	12	1/1	12	Must study	Very likely · 100% Easy
A12 Solid oxide fuel cell (SOFC): favourable characteristics, electrolyte types	12	1/1	12	Must study	Very likely · 100% Easy
A4 A third-generation thin-film cell: dye-sensitised solar cell (working principle)	10	1/1	10	Must study	Very likely · 100% Easy
A7 Fuel cell: definition, difference from a battery, working principle	10	1/1	10	Must study	Very likely · 100% Easy
A10 Molten carbonate fuel cell (MCFC): construction, principle, features	10	1/1	10	Must study	Very likely · 100% Easy
A3 High-efficiency concepts of c-Si cells; the PERL cell	13	1/1	13	Must study	Very likely · 100% Mediun
A6 Multijunction cells: challenges; spectral conversion	13	1/1	13	Must study	Very likely · 100% Mediun
A1 Losses in solar cells; optical losses	10	1/1	10	Must study	Very likely · 100% Mediun

Sort Section B: Best order (likely + easy) | Most likely first | Easiest first | Paper order

SECTION B: SUPERCAPACITORS, HYDROGEN STORAGE, THIN FILMS, XRD AND ELECTRON MICROSCOPY: MARKS ASKED PER PAPER

Topic	2022	Years	Marks	Priority
B10 SEM: backscattered vs secondary electrons; how SEM gives high resolution and depth of field	18	1/1	18	Must study · Very likely · 100% · Easy
B6 Chemical vapour deposition (CVD): process and role in thin films, powders, single crystals; PVD vs CVD	16	1/1	16	Must study · Very likely · 100% · Easy
B5 Sputtering: definition and process	9	1/1	9	Must study · Very likely · 100% · Easy
B8 Geometry and components of an X-ray diffractometer	9	1/1	9	Must study · Very likely · 100% · Easy
B7 Obtaining monochromatic Kα radiation from the X-ray tube spectrum	8	1/1	8	Must study · Very likely · 100% · Easy
B3 Pseudocapacitor vs double-layer capacitor	7	1/1	7	Must study · Very likely · 100% · Easy
B2 Numerical: capacitors 15 F and 8 F in series and in parallel across 50 V	6	1/1	6	Good to have · Very likely · 100% · Easy
B4 Hydrogen storage in carbonaceous materials and CNTs: adsorption mechanism, key parameters, applications	20	1/1	20	Must study · Very likely · 100% · Mediu
B9 XRD numericals: 2θ of Co (111) with Cu Kα; structure factor of an FCC cell	18	1/1	18	Must study · Very likely · 100% · Mediu
B11 TEM: diffraction pattern and image together; preparing a 0.1 mm ceramic sample	17	1/1	17	Must study · Very likely · 100% · Mediu
B1 Electric double-layer capacitor (EDLC): why high capacitance, fundamental principle	12	1/1	12	Must study · Very likely · 100% · Mediu

Section A: Solar cell materials and fuel cells questions 1-4

Sort Section A: Best order (likely + easy) | Most likely first | Easiest first | Paper order

A8 Fuel cell stack (bipolar plates, series connection); DMFC layer structure

Must study · Very likely · 100% · Easy

Asked in 1 of 1 papers · 13 marks in total

STUDY THESE

- One cell gives ~0.7 V, so cells are stacked in series; bipolar plates connect the anode of one cell to the cathode of the next and distribute gases
- Other connections: edge collection (planar), tubular SOFC with interconnects
- DMFC MEA: anode flow plate / diffusion layer / Pt-Ru anode / Nafion membrane / Pt cathode / diffusion layer / cathode plate; methanol + water fed directly

> 2022 Q3(b) 13 marks
>
> How are the individual fuel cells connected within a fuel cell stack? Provide a brief explanation of the layer structure of a Direct Methanol Fuel Cell (DMFC).

Full answer: ESE 4011 Study Pack, topic A8.

A11 Phosphoric acid fuel cell (PAFC): electrolyte properties and functions, electrodes and catalyst

Must study · Very likely · 100% · Easy

Asked in 1 of 1 papers · 13 marks in total

STUDY THESE

- Electrolyte: ~100 % H_3PO_4 held in a SiC-PTFE matrix, 150-220 °C; conducts H^+
- Properties: thermal, chemical, electrochemical stability, low volatility above 150 °C, tolerates CO_2 and ~1 % CO
- Electrodes: PTFE-bonded carbon on carbon paper, Pt (or Pt alloy) catalyst on carbon black, ~0.1-0.5 mg/cm^2

> 2022 Q4(b) 13 marks
>
> What are the properties and functions of phosphoric acid electrolytes in Phosphoric Acid Fuel Cells (PAFCs)? Provide a brief explanation of the electrode and catalyst in PAFCs.

Full answer: ESE 4011 Study Pack, topic A11.

A2 Crystalline silicon structures (mono-, multi-, amorphous); diffusion and drift device concepts

Must study · Very likely · 100% · Easy

Asked in 1 of 1 papers · 12 marks in total

STUDY THESE

- Monocrystalline (one continuous lattice, Cz/FZ), multicrystalline (many grains, cast), amorphous (no long-range order, a-Si:H)
- Diffusion device: thick p-n homojunction; minority carriers reach the junction by diffusion (needs long diffusion length: c-Si)
- Drift device: p-i-n; carriers are swept by the built-in field across the intrinsic absorber (needed when diffusion length is short: a-Si:H)

> 2022 Q1(b) 12 marks
>
> Explain the various structures of crystalline silicon materials used in solar cells. Briefly explain the different structures of a solar cell based on the diffusion and drift device concepts.

Full answer: ESE 4011 Study Pack, topic A2.

A5 Chalcopyrite materials (CIS, CIGS) and the CdTe cell: layer structure and working principle

Must study · Very likely · 100% · Easy

Asked in 1 of 1 papers · 12 marks in total

STUDY THESE

- Chalcopyrites: I-III-VI_2 compounds: $Cu(In,Ga)(Se,S)_2$ - $CuInSe_2$, $CuGaSe_2$, $CuInS_2$, CIGS
- CdTe cell (superstrate): glass / TCO / buffer / n-CdS / p-CdTe / back contact

- Direct band gap 1.45 eV absorbs light in 2 µm; CdS/CdTe heterojunction separates carriers

> 2022 Q2(b) 12 marks
>
> What are the common combinations of elements used to form chalcopyrite materials? Provide a brief explanation of the layer structure of a typical CdTe solar cell and its working principle.

Full answer: ESE 4011 Study Pack, topic A5.

A9 Role of PTFE in alkaline fuel cells; functions of bipolar plates in PEMFCs

Must study · Very likely · 100% · Easy

Asked in 1 of 1 papers · 12 marks in total

STUDY THESE

- PTFE in AFC electrodes: binder and water-repellent (hydrophobic) agent - keeps pores open for gas, stops the liquid KOH electrolyte flooding the electrode, gives a stable three-phase boundary
- Bipolar plates: connect cells electrically in series, distribute fuel and oxidant through flow fields, separate gases, remove water and heat, give mechanical support
- Materials: graphite, coated metals, polymer-carbon composites

> 2022 Q3(c) 12 marks
>
> What is the role of PTFE in Alkaline Fuel Cells (AFCs)? What are the functions of bipolar plates in Proton Exchange Membrane Fuel Cells (PEMFCs)?

Full answer: ESE 4011 Study Pack, topic A9.

A12 Solid oxide fuel cell (SOFC): favourable characteristics, electrolyte types

Must study · Very likely · 100% · Easy

Asked in 1 of 1 papers · 12 marks in total

STUDY THESE

- All-solid ceramic cell at 600-1000 °C; O^{2-} ions move through the electrolyte

- Favourable: high efficiency (and with gas turbine/CHP), fuel flexible with internal reforming, no precious metals, CO is a fuel, no liquid electrolyte management
- Electrolytes: YSZ (yttria-stabilised zirconia), ScSZ, doped ceria (GDC/SDC), LSGM (doped lanthanum gallate), also proton-conducting perovskites

> 2022 Q4(c) 12 marks
>
> Explain the favorable characteristics of a Solid Oxide Fuel Cell (SOFC). Explain the types of electrolytes used in this fuel cell.

Full answer: ESE 4011 Study Pack, topic A12.

A4 A third-generation thin-film cell: dye-sensitised solar cell (working principle)

Must study · Very likely · 100% · Easy

Asked in 1 of 1 papers · 10 marks in total

STUDY THESE

- Third generation = low-cost or beyond-Shockley-Queisser concepts: DSSC, organic, perovskite, quantum-dot, multijunction
- DSSC: dye on mesoporous TiO_2 absorbs light, injects electron into TiO_2; iodide electrolyte regenerates the dye; triiodide reduced at Pt counter electrode
- Light absorption and charge transport are done by different materials (unlike Si)

> 2022 Q2(a) 10 marks
>
> Mention a third generation solar cell belonging to the group of thin film solar cells. Explain the working principle of such a solar cell with a suitable diagram.

Full answer: ESE 4011 Study Pack, topic A4.

A7 Fuel cell: definition, difference from a battery, working principle

Must study · Very likely · 100% · Easy

Asked in 1 of 1 papers · 10 marks in total

STUDY THESE

- Fuel cell = electrochemical device converting chemical energy of a fuel (H_2) and oxidant (O_2) directly into electricity, as long as fuel is supplied

- Battery stores its reactants inside and runs down/needs recharging; fuel cell is an energy converter with external fuel
- $H_2 \rightarrow 2H^+ + 2e^-$ at anode; ions through electrolyte, electrons through external load; $O_2 + H^+ + e^- \rightarrow H_2O$ at cathode

> 2022 Q3(a) 10 marks
>
> What is a fuel cell and how does it differ from a conventional battery? Also briefly explain the working principle of a fuel cell.

Full answer: ESE 4011 Study Pack, topic A7.

A10 Molten carbonate fuel cell (MCFC): construction, principle, features

Must study · Very likely · 100% · Easy

Asked in 1 of 1 papers · 10 marks in total

STUDY THESE

- Electrolyte: molten Li/K carbonate in a $LiAlO_2$ matrix at about 650 °C; ion = CO_3^{2-}
- Anode Ni-Cr/Ni-Al, cathode lithiated NiO; CO_2 must be recycled from anode to cathode
- Features: no precious metal, internal reforming, fuel flexible (CO is a fuel), high efficiency with heat recovery; corrosion and slow start

> 2022 Q4(a) 10 marks
>
> Briefly explain the construction and operating principle of Molten Carbonate Fuel Cells (MCFCs). Also what are the general features of MCFCs?

Full answer: ESE 4011 Study Pack, topic A10.

A3 High-efficiency concepts of c-Si cells; the PERL cell

Must study · Very likely · 100% · Medium

Asked in 1 of 1 papers · 13 marks in total

STUDY THESE

- High-efficiency concepts reduce optical, recombination and resistive losses: passivation, light trapping, local contacts, selective emitters

- PERL = passivated emitter, rear locally-diffused: inverted pyramids + ARC, thin oxide passivation front and rear, rear point contacts with local p+ diffusion
- Record about 25 % (UNSW); parent of PERC/PERT/TOPCon industrial cells

> 2022 Q1(c) 13 marks
>
> What is meant by high-efficiency concepts of solar cells? Explain with a proper schematic diagram how the PERL concept contributes to the overall advancement of crystalline silicon solar cell technology.

Full answer: ESE 4011 Study Pack, topic A3.

A6 Multijunction cells: challenges; spectral conversion

Must study · Very likely · 100% · Medium

Asked in 1 of 1 papers · 13 marks in total

STUDY THESE

- Multijunction: stack of cells with decreasing band gaps, each absorbing part of the spectrum (efficiency > 40 % with concentration)
- Challenges: current matching, lattice matching, tunnel junctions, cost, spectral sensitivity, thermal mismatch
- Spectral conversion: down-conversion (1 high-energy photon → 2 lower), down-shifting, up-conversion (2 sub-gap photons → 1 usable) to reduce thermalisation and transmission losses

> 2022 Q2(c) 13 marks
>
> Briefly explain the challenges associated with multijunction solar cells. What is spectral conversion in the context of solar cells, and how does it enhance their efficiency?

Full answer: ESE 4011 Study Pack, topic A6.

A1 Losses in solar cells; optical losses

Must study · Very likely · 100% · Medium

Asked in 1 of 1 papers · 10 marks in total

STUDY THESE

- Losses: optical (reflection, shading, incomplete absorption), thermalisation and non-absorption (spectral mismatch), recombination, resistive (series and shunt)

- Optical losses = light that never creates an electron-hole pair: reflected from the surface, blocked by front metal, or passing through without absorption
- Reduced by anti-reflection coating, surface texturing, thin fingers, back reflectors (light trapping)

> 2022 Q1(a) 10 marks
>
> What are the different types of losses encountered in solar cells? Explain the concept of optical losses in solar cells with the proper schematic diagram.

Full answer: ESE 4011 Study Pack, topic A1.

Section B: Supercapacitors, hydrogen storage, thin films, XRD and electron microscopy

questions 5-8

Sort Section B: Best order (likely + easy) | Most likely first | Easiest first | Paper order

B10 SEM: backscattered vs secondary electrons; how SEM gives high resolution and depth of field

Must study · Very likely · 100% · Easy

Asked in 1 of 1 papers · 18 marks in total

STUDY THESE

- SE: low-energy (< 50 eV) electrons knocked out of specimen atoms near the surface (inelastic) - topography, best resolution
- BSE: beam electrons elastically scattered back (> 50 eV, near beam energy), from deeper - atomic-number (Z) contrast and composition
- SEM: fine focused beam scanned in a raster; small probe gives resolution (~1-10 nm), small aperture and long working distance give large depth of field (~100× optical microscope)

> 2022 Q8(a) 6 marks
>
> Differentiate between backscattered electrons and secondary electrons in terms of their formation and role during image generation in SEM.

> 2022 Q8(b) 12 marks
>
> How does Scanning Electron Microscopy (SEM) enable the examination of microscopic structures with high resolution and depth of field?

Full answer: ESE 4011 Study Pack, topic B10.

B6 Chemical vapour deposition (CVD): process and role in thin films, powders, single crystals; PVD vs CVD

Must study · Very likely · 100% · Easy

Asked in 1 of 1 papers · 16 marks in total

STUDY THESE

- CVD: gaseous precursors react/decompose on or near a heated substrate to form a solid deposit; steps: transport, adsorption, reaction, nucleation/growth, desorption of by-products
- Thin films (Si, SiO_2, SiN, diamond, graphene), powders (gas-phase nucleation: fumed silica, nanoparticles, CNTs), single crystals (epitaxy, bulk SiC crystals)
- PVD: physical (evaporation, sputtering), line-of-sight, lower temperature. CVD: chemical reaction, conformal coverage, higher temperature, toxic precursors

> 2022 Q6(c) 10 marks
>
> Discuss the process of Chemical Vapor Deposition (CVD) and its role in the fabrication of thin films, powders, and single crystals.

> 2022 Q6(d) 6 marks
>
> Compare and contrast between PVD and CVD process.

Full answer: ESE 4011 Study Pack, topic B6.

B5 Sputtering: definition and process

Must study · Very likely · 100% · Easy

Asked in 1 of 1 papers · 9 marks in total

STUDY THESE

- Sputtering = ejection of atoms from a solid target by bombardment with energetic ions (usually Ar^+), the ejected atoms condensing on a substrate as a thin film
- Glow discharge plasma between target (cathode) and substrate (anode) in Ar at low pressure; DC for metals, RF for insulators, magnetron for higher rate, reactive

sputtering for oxides/nitrides
- Momentum transfer (collision cascade), not evaporation

> 2022 Q6(b) 9 marks
>
> Define sputtering. Describe the sputtering process with a suitable figure.

Full answer: ESE 4011 Study Pack, topic B5.

B8 Geometry and components of an X-ray diffractometer

Must study · Very likely · 100% · Easy

Asked in 1 of 1 papers · 9 marks in total

STUDY THESE

- Bragg-Brentano θ-2θ geometry: X-ray tube, divergence slit, flat specimen at centre, receiving slit, detector on the diffractometer circle
- Specimen turns θ while detector turns 2θ (goniometer 1:2 coupling); focusing circle keeps the beam focused
- Soller slits limit divergence; filter or monochromator removes Kβ; detector records intensity vs 2θ

> 2022 Q7(b) 9 marks
>
> Draw a geometric arrangement of an X-ray diffractometer with the functions of each component.

Full answer: ESE 4011 Study Pack, topic B8.

B7 Obtaining monochromatic Kα radiation from the X-ray tube spectrum

Must study · Very likely · 100% · Easy

Asked in 1 of 1 papers · 8 marks in total

STUDY THESE

- Tube gives continuous (white) radiation plus characteristic Kα and Kβ lines
- Filter: a foil whose K absorption edge lies between Kβ and Kα of the target (Ni for Cu, Zr for Mo) absorbs Kβ strongly and passes Kα
- Better: crystal monochromator (diffracting crystal set at Bragg angle for Kα) and pulse-height discrimination in the detector; choose tube voltage below excess

2022 Q7(a) 8 marks

Explain how you will obtain monochromatic Kα radiation from the continuous X-rays and other characteristic X-rays.

Full answer: ESE 4011 Study Pack, topic B7.

B3 Pseudocapacitor vs double-layer capacitor

Must study · Very likely · 100% · Easy

Asked in 1 of 1 papers · 7 marks in total

STUDY THESE

- Pseudocapacitance = Faradaic charge storage: fast, reversible redox reactions, intercalation or electrosorption at/near the electrode surface
- Materials: RuO_2, MnO_2, other transition-metal oxides, conducting polymers (PANI, PPy, PEDOT)
- Higher capacitance and energy than EDLC but lower power and cycle life

2022 Q5(c) 7 marks

"Pseudocapacitor is a kind of supercapacitor which is different from double layer capacitor" - explain why.

Full answer: ESE 4011 Study Pack, topic B3.

B2 Numerical: capacitors 15 F and 8 F in series and in parallel across 50 V

Good to have · Very likely · 100% · Easy

Asked in 1 of 1 papers · 6 marks in total

STUDY THESE

- Series: same charge Q; $C_{eq} = C_1C_2/(C_1 + C_2)$; $V_i = Q/C_i$ (smaller C takes larger voltage)
- Parallel: same voltage across both
- Answer: series 17.39 V and 32.61 V; parallel 50 V each

2022 Q5(b) 6 marks

If two capacitors C_1 = 15 F and C_2 = 8 F are connected in series and in parallel to a 50 V battery, determine the voltage across each capacitor for each type of

connection.

Full answer: ESE 4011 Study Pack, topic B2.

B4 Hydrogen storage in carbonaceous materials and CNTs: adsorption mechanism, key parameters, applications

Must study · Very likely · 100% · Medium

Asked in 1 of 1 papers · 20 marks in total

STUDY THESE

- Physisorption: H_2 molecules held by weak van der Waals forces on carbon surfaces (~4-6 kJ/mol), significant only at low temperature (77 K) and high pressure; reversible
- Also chemisorption (H atoms bond to C, e.g. via spillover from metal catalysts) - stronger but harder to release
- Increase capacity: high specific surface area, optimum pore size (~0.6-0.7 nm), opening tube ends, purification, defects, doping/metal decoration (Pd, Pt, Li, Ti), functionalisation, low temperature/high pressure

> 2022 Q5(d) 10 marks
> Describe the hydrogen adsorption mechanism in carbonaceous materials. What are the key parameters for increasing H_2 storage capacity in CNTs?

> 2022 Q6(a) 10 marks
> How do carbonaceous materials such as carbon nanotubes contribute to hydrogen storage and what are their potential applications in this field?

Full answer: ESE 4011 Study Pack, topic B4.

B9 XRD numericals: 2θ of Co (111) with Cu Kα; structure factor of an FCC cell

Must study · Very likely · 100% · Medium

Asked in 1 of 1 papers · 18 marks in total

STUDY THESE

- Cubic: $d_{hkl} = a / \sqrt{(h^2 + k^2 + l^2)}$; Bragg: $\lambda = 2d \sin\theta$

- Co (111): d = 2.046 Å, θ = 22.13°, 2θ = 44.27°
- FCC: F = 4f if h, k, l all even or all odd (unmixed), F = 0 if mixed: (111), (200), (220), (311) reflect; (100), (110), (210) do not

> 2022 Q7(c) 6 marks
>
> A crystal of cobalt (Co) is analysed using CuKα (λ = 1.5418 Å) X-ray radiation. Calculate the angle of reflection 2θ of the (1 1 1) plane. The lattice parameter of FCC cobalt is 3.544 Å.

> 2022 Q7(d) 12 marks
>
> Calculate the structure factor for an FCC unit cell and show six planes from where reflection may occur or not.

Full answer: ESE 4011 Study Pack, topic B9.

B11 TEM: diffraction pattern and image together; preparing a 0.1 mm ceramic sample

Must study · Very likely · 100% · Medium

Asked in 1 of 1 papers · 17 marks in total

STUDY THESE

- The objective lens forms a diffraction pattern in its back focal plane and an image in its image plane at the same time; the intermediate lens chooses which one is magnified onto the screen
- Selected-area aperture in the image plane picks the area for diffraction (SAED); objective aperture in the back focal plane picks beams for bright/dark-field images
- Ceramic sample prep: cut 3 mm disc, grind/polish to ~100 µm (given), dimple grind centre to ~10-20 µm, Ar ion mill to perforation (electron transparent < 100 nm), carbon coat to stop charging; or FIB lift-out / tripod polishing

> 2022 Q8(c) 8 marks
>
> How are diffraction pattern and image simultaneously present in TEM?

> 2022 Q8(d) 9 marks
>
> You need to analyse a ceramic sample (0.1 mm thick) in TEM. Explain how you will prepare the sample for TEM investigation.

Full answer: ESE 4011 Study Pack, topic B11.

B1 Electric double-layer capacitor (EDLC): why high capacitance, fundamental principle

Must study · Very likely · 100% · Medium

Asked in 1 of 1 papers · 12 marks in total

STUDY THESE

- C = εA/d: EDLC has huge area (activated carbon 1000-3000 m^2/g) and tiny charge separation d (~0.3-1 nm double layer), so capacitance is thousands of farads
- Charge is stored electrostatically by ions forming a Helmholtz double layer at each electrode; no reaction, so very fast and long cycle life
- Two double layers in series: $1/C = 1/C_+ + 1/C_-$; energy $E = ½CV^2$

> 2022 Q5(a) 12 marks
>
> Why does an Electric Double Layer Capacitor (EDLC) show high capacitance? Describe the fundamental principles of EDLC with necessary figure.

Full answer: ESE 4011 Study Pack, topic B1.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
µc-Si:H	hydrogenated microcrystalline silicon
a-Si:H	hydrogenated amorphous silicon
AFC	alkaline fuel cell
ALD	atomic layer deposition
APCVD	atmospheric-pressure CVD
ARC	anti-reflection coating
BSE	backscattered electrons
c-Si	crystalline silicon
CGS	copper gallium selenide
CHP	combined heat and power

CIGS	copper indium gallium (di)selenide, $Cu(In,Ga)Se_2$
CIS	copper indium (di)selenide, $CuInSe_2$
CNT	carbon nanotube
CO	carbon monoxide
CV	cyclic voltammetry
CVD	chemical vapour deposition
Cz	Czochralski (crystal pulling)
CZTS	copper zinc tin sulphide (kesterite)
DC	direct current
DMFC	direct methanol fuel cell
DOE	US Department of Energy
DSSC	dye-sensitised solar cell
EBSD	electron backscatter diffraction
EDLC	electric double-layer capacitor
EDS	energy-dispersive X-ray spectroscopy
EDX	energy-dispersive X-ray (analysis)
Eg	band gap energy
ESR	equivalent series resistance
eV	electron-volt (energy unit, 1.6×10^{-19} J)
FEG	field-emission gun
FIB	focused ion beam
FTO	fluorine-doped tin oxide
FZ	float zone (refining)
GDC	gadolinium-doped ceria
GDL	gas diffusion layer
HIT	heterojunction with intrinsic thin layer
HRTEM	high-resolution TEM
IBC	interdigitated back contact
ICDD	International Centre for Diffraction Data

IR	infrared
ITO	indium tin oxide
IUPAC	International Union of Pure and Applied Chemistry
JCPDS	Joint Committee on Powder Diffraction Standards
LPCVD	low-pressure CVD
LSCF	lanthanum strontium cobalt ferrite
LSGM	strontium- and magnesium-doped lanthanum gallate
LSM	strontium-doped lanthanum manganite
mc-Si	multicrystalline silicon
MCFC	molten carbonate fuel cell
MEA	membrane electrode assembly
MOCVD	metal-organic CVD
MOF	metal-organic framework
MWCNT	multi-walled carbon nanotube
PAFC	phosphoric acid fuel cell
PANI	polyaniline
PCFC	proton-conducting ceramic fuel cell
PECVD	plasma-enhanced chemical vapour deposition
PEDOT	poly(3,4-ethylenedioxythiophene)
PEM	proton exchange (polymer electrolyte) membrane
PEMFC	proton exchange membrane fuel cell
PERC	passivated emitter and rear cell
PERL	passivated emitter, rear locally-diffused (cell)
PERT	passivated emitter, rear totally-diffused
PLD	pulsed laser deposition
PPy	polypyrrole
PTFE	polytetrafluoroethylene (Teflon)
PV	photovoltaic
PVD	physical vapour deposition

RF	radio frequency
SAED	selected-area electron diffraction
ScSZ	scandia-stabilised zirconia
SDC	samarium-doped ceria
SE	secondary electrons
SEM	scanning electron microscope
SHJ	silicon heterojunction
SOEC	solid oxide electrolysis cell
SOFC	solid oxide fuel cell
SWCNT	single-walled carbon nanotube
TCO	transparent conducting oxide
TEM	transmission electron microscope
TOPCon	tunnel oxide passivated contact
UNSW	University of New South Wales
UV	ultraviolet
wt %	percent by weight
XRD	X-ray diffraction
YSZ	yttria-stabilised zirconia

Built from the scanned ESE 4011 papers (2022). Question text is cleaned up from the scans.